

Comparison of Geometric and Energy-Based Error Scaling for Recursive Intrinsic Multiscale Filtering

1 Task

The notebook [gd_imf_real_vs_calculated.ipynb](#) scales every stage- k error by the geometric factor $a^{k/2}$, where $a = \sqrt{2}$ is the bandwidth reduction factor. The companion notebook [gd_imf_real_vs_calculated.energy](#) stored with the other real-vs-calculated experiments, uses a stage-specific error scale derived from the energy of the complete recursive filter. For the nonlinear robust decomposition, the corresponding scale is estimated by a parametric bootstrap.

The methods answer different questions:

- *Geometric scaling*: does the error follow the theoretical bandwidth order $(nh_k)^{-1/2}$?
- *Energy scaling*: is the error consistent with the variance of the complete stage-specific operator, including all previous residual subtractions?

The comparison should therefore consider calibration, assumptions, and computational cost in addition to visual flatness.

2 Common Error Quantities

Let W_k denote the local linear smoother at IMF stage k , and define the residual operator

$$A_{k-1} = (I - W_{k-1}) \cdots (I - W_1), \quad A_0 = I.$$

The complete recursive component operator is

$$\mathcal{W}_k = W_k A_{k-1}.$$

For an observation $Y = X + \varepsilon$, the linear recursive component error is

$$e_k = \tilde{S}^{(k)} - S^{(k)} = \mathcal{W}_k \varepsilon.$$

The first component is low-pass because $A_0 = I$. Later recursive components are band-pass because A_{k-1} removes the low-frequency content extracted at the previous stages.

The main scalar diagnostics are

$$\text{RMSE}(e_k) = \left\{ \frac{1}{n} \sum_{t=1}^n e_k(t)^2 \right\}^{1/2}, \quad \|e_k\|_\infty = \max_t |e_k(t)|.$$

3 Method 1: Geometric Bandwidth Scaling

The bandwidth schedule is approximately

$$h_k = \frac{h_1}{a^{k-1}}, \quad a = \sqrt{2}.$$

The original notebook defines

$$e_k^{\text{geo}} = \frac{e_k}{a^{k/2}}.$$

The natural stage-one convention would use $a^{(k-1)/2}$, but the two choices differ by the same constant $a^{1/2}$ at every stage. The indexing convention cannot create or remove a stage-one discontinuity.

The variance order $1/(nh)$ is standard in kernel and local-polynomial regression; see, for example, Fan and Gijbels [3] and Tsybakov [4]. The underlying kernel estimator was introduced independently by Nadaraya [1] and Watson [2]. In Nadaraya's note, Theorem 1 (starting on p. 141 and continuing at the top of p. 142) gives the random-design $\sqrt{nh}$ limit and displays the corresponding kernel-energy factor $\int K^2$. The present experiment instead has a regular fixed design. For that design, a normalized kernel average with bandwidth h has the asymptotic variance

$$\text{Var}\{\tilde{S}_h(t) - S_h(t)\} \approx \frac{\sigma^2}{nh} \frac{\int K(u)^2 du}{\left\{\int K(u) du\right\}^2}.$$

This expression assumes that the number of observations in a window grows, $nh \rightarrow \infty$, and that the noise values have a common variance and are uncorrelated. The design-density term that appears in the random-design formula is constant here and is absorbed into the multiplicative constant.

For a recursive IMF component, the relevant operator is not one kernel average but $\mathcal{W}_k = W_k A_{k-1}$. Lemma 2.5 of the supplied IMF draft [5] states the corresponding order

$$\text{Var}\{e_k(t)\} \asymp \frac{\sigma^2}{nh_k}.$$

Here $\asymp$ means that the two sides have the same order up to positive multiplicative constants; it does not mean equality. Writing the stage-specific constant explicitly gives

$$\text{Var}\{e_k(t)\} \approx \frac{\sigma^2}{nh_k} C_k.$$

The constant C_k contains the squared energy of the complete recursive filter, including the previous residual operators. This is the recursive analogue of the kernel-dependent constant in the Nadaraya–Watson variance formula. Geometric scaling compensates for $h_k^{-1/2}$ but leaves $\sqrt{C_k}$ in the scaled error, so it checks the bandwidth rate rather than equality of stage constants.

For the current recursive filters, C_1 is about 0.90, while the later constants are about 0.09. A factor of approximately

$$\sqrt{0.90/0.09} = \sqrt{10} \approx 3.16$$

can remain after geometric scaling without contradicting the common bandwidth order.

3.1 Why the first recursive operator has greater energy

The difference is a consequence of the initialization of the recursion, rather than of the bandwidth schedule alone. On the periodic regular grid used in the notebooks, the smoothers are circulant.

Let $w_k(\omega_m)$ be the discrete Fourier multiplier of W_k . The multiplier and squared row energy of the complete stage operator are

$$q_k(\omega_m) = w_k(\omega_m) \prod_{j < k} \{1 - w_j(\omega_m)\}, \quad E_k = \|\ell_k\|_2^2 = \frac{1}{n} \sum_m |q_k(\omega_m)|^2.$$

Thus operator energy is the integrated squared gain: it measures how much white noise power the entire stage transmits. More precisely, the stage-one spike after geometric scaling is controlled by the bandwidth-normalized energy $C_k = (nh_k)E_k$. It is this constant, rather than necessarily the unnormalized E_k for every increasingly narrow later filter, that is exceptionally large at stage one.

For $k = 1$, the product is empty, so $q_1 = w_1$. The first stage therefore keeps the full low-frequency passband of a normalized low-pass smoother. In particular, $w_1(0) = 1$, and $|w_1(\omega)|$ remains close to one throughout a neighborhood of zero frequency. It is this whole low-frequency neighborhood, not the single DC Fourier coefficient by itself, that contributes the large energy.

For every $k > 1$, at least one residual factor is present. Since every normalized smoother preserves constants, $w_j(0) = 1$, and hence

$$q_k(0) = w_k(0) \prod_{j < k} \{1 - w_j(0)\} = 0.$$

More generally, wherever an earlier low-pass multiplier $w_j(\omega)$ is close to one, the factor $1 - w_j(\omega)$ is close to zero. The earlier residual operators therefore remove the low-frequency region in which the current smoother would otherwise have its largest gain. At the other end of the spectrum, $w_k(\omega)$ is already small. The product can consequently transmit mainly the narrower transition region between these two regimes: later components are band-pass details, rather than low-pass averages of the original data. The first residual factor creates most of the drop, and the older factors provide further suppression.

The same distinction appears in the equivalent-kernel constant. If $L = K / \int K$ and $h_j/h_k \approx a^{k-j}$, then, up to the Fourier convention,

$$C_k = \frac{1}{2\pi} \int |\widehat{L}(z)|^2 \prod_{\ell=1}^{k-1} |1 - \widehat{L}(a^\ell z)|^2 dz.$$

At stage one there are no attenuating factors. For the implemented kernel $K(u) = \frac{3}{4}(1 - |u|)_+^2$,

$$C_1 = \int \left(\frac{K(u)}{\int K} \right)^2 du = 0.9.$$

In the exact finite- n calculation using the notebook's discrete windows, $C_k = (nh_k)E_k$ equals 0.9009 at stage 1, 0.1518 at stage 2, 0.1090 at stage 3, and approximately 0.09 thereafter. Equivalently, the residual product leaves 16.8% of the corresponding single-pass variance at stage 2 and about 9.7%–12.1% at later stages. This calculation is specific to the implemented kernel, window schedule, and periodic boundary; the qualitative low-pass-versus-band-pass mechanism is general, but a tenfold energy ratio is not a universal IMF constant.

For the robust recursion there is no single fixed global Fourier multiplier. Nevertheless, the local Jacobian of a converged normalized robust smoother has rows summing to one, so its first-stage linearization preserves constants while the Jacobian of each residual map removes them. This gives the same low-pass-versus-band-pass explanation locally. The robust energy ratio in this document is therefore supported by the bootstrap experiment, rather than by the exact linear identity above.

Geometric scaling is useful when the goal is to check the predicted bandwidth rate with a simple deterministic normalization. It should not be expected to make the recursive error curve flat, because it does not remove C_k and hence does not account for the difference between the first low-pass component and the later band-pass components. The same rate also does not calibrate a sup error.

4 Method 2: Stage-Specific Energy Scaling

4.1 Exact linear scaling

Assume that every noise value has mean zero and the same variance σ^2 , and that different noise values are uncorrelated:

$$\text{Var}(\varepsilon_t) = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_u) = 0 \quad (t \neq u).$$

Let $\ell_{k,q}$ be the effective weight placed on the noise value at offset q by the complete recursive stage. In other words,

$$e_k(t) = \sum_q \ell_{k,q} \varepsilon_{t+q}.$$

With the notebook's wrap boundary, the same weights are shifted around the time grid at every t . It follows directly that

$$\text{Var}\{e_k(t)\} = \sigma^2 (\mathcal{W}_k \mathcal{W}_k^\top)_{tt} = \sigma^2 \sum_q \ell_{k,q}^2.$$

The exact recursive energy scale is

$$s_{k,\text{rec}} = \sigma \left(\sum_q \ell_{k,q}^2 \right)^{1/2},$$

and the standardized error is

$$z_{k,\text{rec}} = \frac{e_k}{s_{k,\text{rec}}}.$$

For a periodic regular design, all rows of $\mathcal{W}_k$ have the same norm, so

$$\text{E}[\text{RMSE}(z_{k,\text{rec}})^2] = 1.$$

This normalization removes both the bandwidth factor and the stage-specific operator-energy constant.

4.2 Bootstrap robust scaling

The robust smoother is nonlinear, so there is no fixed global matrix $\mathcal{W}_k$. The companion notebook estimates the scale under the same clean signal, Gaussian noise, signed contamination law, robust loss, window schedule, and clean-fit reference as the displayed experiment.

For bootstrap replicate $b = 1, \dots, B$, let $e_k^{(b)}(t)$ be the robust recursive error. The estimated scale is

$$\hat{s}_{k,\text{rec}} = \left\{ \frac{1}{B} \sum_{b=1}^B \frac{1}{n} \sum_{t=1}^n \left(e_k^{(b)}(t) \right)^2 \right\}^{1/2}.$$

The displayed error is standardized as

$$\hat{z}_{k,\text{rec}} = \frac{e_k}{\hat{s}_{k,\text{rec}}}.$$

The current notebook uses $B = 32$ deterministic replicates for a quick reproducible analysis. For publication-level scale estimates, a larger value such as $B \geq 200$ and a Monte Carlo uncertainty assessment are preferable.

Energy scaling is intended as a calibration rather than only a rate check. In the linear case it gives an exact unit target for the expected squared standardized RMSE. In the robust case it is necessarily model-dependent: the result changes with the signal, contamination law, robust-loss parameter, reference definition, and bootstrap size. It is therefore more expensive and less universal than the geometric normalization, and its RMS scale still does not calibrate sup errors.

5 Relationship Between the Methods

If

$$s_k^2 \approx \frac{\sigma^2}{nh_k} C_k,$$

then, because $h_k \approx h_1 a^{-(k-1)}$,

$$\frac{e_k}{a^{k/2}} \text{ retains a factor proportional to } \sqrt{C_k},$$

whereas

$$\frac{e_k}{s_k}$$

removes both the bandwidth rate and C_k . Thus the methods are complementary: the first tests theoretical order, while the second tests stage-specific calibration.

Criterion	Geometric scaling	Energy scaling
Primary question	Does error follow the bandwidth rate?	Is error calibrated to the complete stage operator?
Denominator	$a^{k/2}$	$\sigma \ \ell_k\ _2$ or bootstrap estimate
Stage-dependent constant	Remains in the result	Included in the denominator
Stage 1	Compared directly with later details	Assigned its own low-pass energy scale
Linear computation	Constant time once k is known	Exact FFT/impulse-response calculation
Robust computation	No additional computation	Parametric bootstrap or Jacobian/sandwich approximation
Expected calibrated RMSE	No unit target	Approximately one
Sup-error interpretation	Not calibrated	Still not calibrated by an RMS scale
Model dependence	Low	Higher, especially for robust bootstrap

Table 1: Conceptual comparison of the two error normalizations.

6 Empirical Comparison

Table 2 gives the recursive RMSE series under both methods. The columns have different units: geometric values retain the units of the original signal up to the dimensionless bandwidth factor, while energy-scaled values are dimensionless.

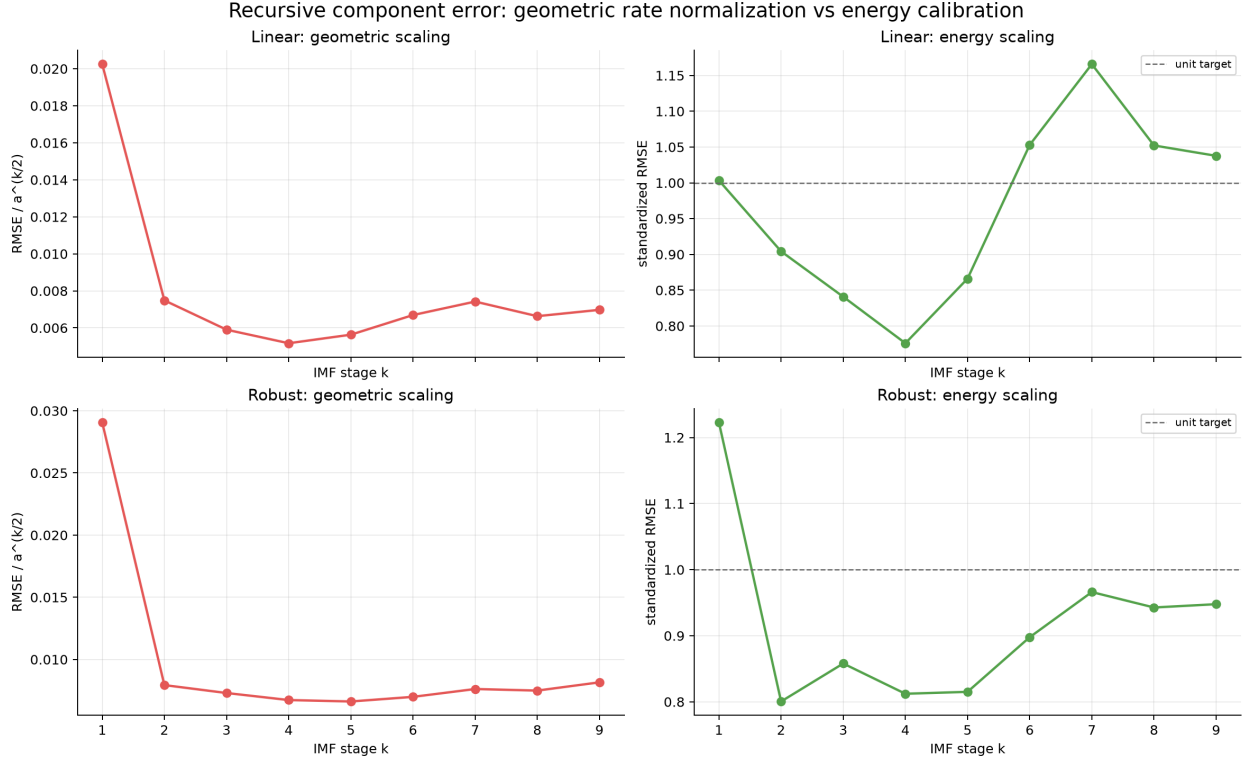


Figure 1: Recursive component-error RMSE under geometric bandwidth scaling and stage-specific energy scaling. The dashed line in each energy panel is the unit calibration target. The geometric panels retain the stage-dependent filter-energy constant, while the energy panels divide by it.

Stage	Linear geometric	Linear energy	Robust geometric	Robust energy
1	0.020249	1.003	0.029067	1.223
2	0.007485	0.904	0.007936	0.800
3	0.005903	0.841	0.007300	0.858
4	0.005170	0.776	0.006733	0.812
5	0.005631	0.866	0.006619	0.815
6	0.006692	1.053	0.006990	0.898
7	0.007421	1.166	0.007622	0.966
8	0.006632	1.052	0.007494	0.943
9	0.006974	1.038	0.008159	0.948

Table 2: Recursive RMSE after geometric or energy scaling.

The stage-one-to-tail ratio is defined by

$$R_{1:\text{tail}} = \frac{r_1}{\frac{1}{8} \sum_{k=2}^9 r_k},$$

where r_k is the scaled recursive RMSE. Flatness is summarized by the sample coefficient of variation

$$CV(r_1, \dots, r_9) = \frac{\text{sd}(r_1, \dots, r_9)}{\text{mean}(r_1, \dots, r_9)}.$$

In this document, “flat” has a narrow meaning: the scaled recursive RMSE is approximately unchanged as the stage index k changes. It does not mean that the raw errors are small or that the recovered components are accurate. A perfectly flat nine-stage sequence would have $R_{1:\text{tail}} = 1$ and $CV = 0$. Thus $|R_{1:\text{tail}} - 1|$ measures the special mismatch of stage 1 relative to stages 2–9, while the CV measures variation over all nine stages. Values closer to zero in Figure 2 mean that one scaling rule gives the stages a more uniform error scale.

Method	Case	$R_{1:\text{tail}}$	CV, stages 1–9	CV, stages 2–9
Geometric	Linear recursive	3.121	58.0%	13.0%
Energy	Linear recursive	1.043	13.0%	13.9%
Geometric	Robust recursive	3.951	74.3%	7.5%
Energy	Robust recursive	1.390	14.2%	7.7%

Table 3: Numerical ingredients of the flatness comparison. The ideal values are $R_{1:\text{tail}} = 1$ and $CV = 0$.

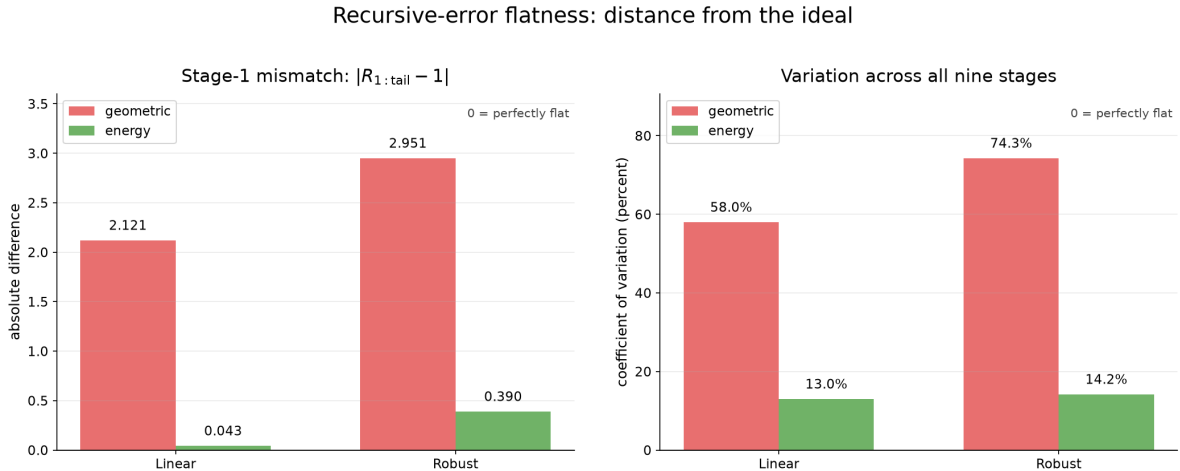


Figure 2: Recursive error flatness. The left panel plots $|R_{1:\text{tail}} - 1|$, so zero means that stage 1 has the same scaled RMSE as the mean of stages 2–9. The right panel plots the coefficient of variation over all nine scaled RMSE values, where zero means that every stage has exactly the same value. Smaller bars indicate a more uniform stage-to-stage scale, not smaller raw errors.

For the linear recursive series, energy scaling changes the stage-one-to-tail ratio from 3.121 to 1.043. For the robust series, it changes the ratio from 3.951 to 1.390. The remaining robust difference is plausible for one fixed noise realization and a scale estimated from only 32 bootstrap replicates; it should be assessed across repeated observations rather than forced to equal one.

The tail CV is not materially reduced because stages 2–9 were already close to a common lower plateau under geometric scaling. The main benefit of energy scaling is therefore not generic smoothing of the curve: it corrects the structural mismatch between the first low-pass component and the later band-pass components.

7 Recommended Comparison Protocol

A fair evaluation should use the same raw component errors and apply both normalizations afterward.

1. Show both the geometric rate-normalized recursive error and the energy-standardized error; do not silently replace one interpretation with the other.
2. Report results for all stages and for the tail stages $k \geq 2$ or $k \geq 3$. This separates initialization effects from steady multiscale behavior.
3. For each method, report:
 - stage-one-to-tail ratio;
 - all-stage and tail-stage coefficient of variation;
 - mean and range of standardized RMSE;
 - computational time and, for robust scaling, bootstrap uncertainty.
4. Repeat the experiment across noise seeds. For energy scaling, check whether $E[\text{RMSE}(z_k)^2]$ is close to one, not whether every realized $\text{RMSE}(z_k)$ equals one.
5. Compare linear and robust methods on matched Gaussian-only and contaminated observations if the goal is method comparison.
6. Evaluate sup errors separately. A suitable comparison can use stage-specific bootstrap quantiles of $\|e_k\|_\infty$ or a simultaneous confidence-band normalization.

8 Recommendation

Both methods should be retained because they diagnose different properties.

- Use geometric scaling as a compact theoretical-rate diagnostic and to compare with statements of order $(nh_k)^{-1/2}$.
- Use energy scaling as the primary stage-to-stage calibration for recursive IMF errors. It correctly accounts for the low-pass first component and the lower-energy band-pass components that follow.
- For the robust decomposition, increase the bootstrap size and report Monte Carlo uncertainty before using the energy scales for formal inference.
- Do not interpret either RMS normalization as a proof of constant sup error.

The clearest presentation is a side-by-side pair of plots: geometric scaling on the left, labeled “bandwidth-order normalization,” and energy scaling on the right, labeled “stage-specific RMS calibration.”

9 Reproducibility Files

- Original geometric-scaling notebook: [gd_imf_real_vs_calculated.ipynb](#).
- Energy-scaled reproduction: [gd_imf_real_vs_calculated_energy_scaled.ipynb](#).
- Investigation summary: [research/first-imf-recursive-error/executive-summary.md](#).
- Detailed theory review: [research/first-imf-recursive-error/agents/theory.md](#).

References

- [1] E. A. Nadaraya. On estimating regression. *Theory of Probability and Its Applications*, 9(1):141–142, 1964. [doi:10.1137/1109020](#); local copy: [nadaraya1964.pdf](#).
- [2] G. S. Watson. Smooth regression analysis. *Sankhyā: The Indian Journal of Statistics, Series A*, 26(4):359–372, 1964. [JSTOR 25049340](#); [DTIC technical-report record AD0467119](#).
- [3] J. Fan and I. Gijbels. *Local Polynomial Modelling and Its Applications*. Chapman & Hall, London, 1996.
- [4] A. B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, New York, 2009. [doi:10.1007/b13794](#).
- [5] V. Spokoiny. *Intrinsic robust multiscale filtering*. Unpublished project draft, July 3, 2026, especially Lemma 2.5.